

Scenario

A laboratory system has been set up to return an XML formatted laboratory message over an HTTP connection. Messages are retrieved from the HTTP server by issuing a GET request on the following URL:

<http://resources.edu.rhapsody.global/professional/assessment/getReports?user=<email>>

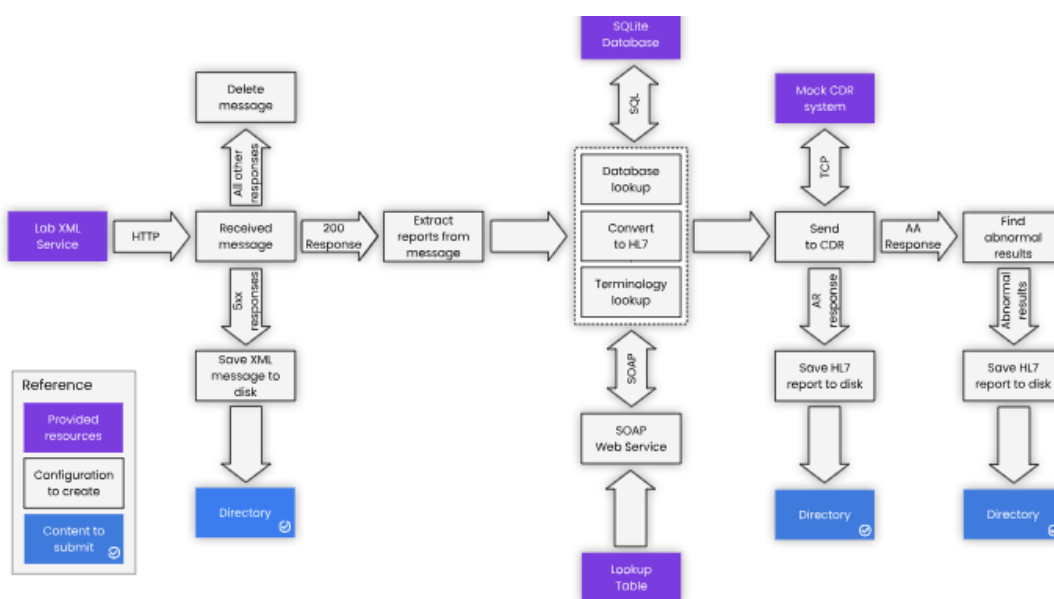
Where <email> is the email address associated with your Academy account.

If there are laboratory messages available for retrieval, the server will respond with an HTTP status code of 200 and send the XML formatted message in the response body. Where multiple reports are available, the message will contain details for all reports. If no message exists, the server will respond with a status code of 404. Errors on the server will be reported with either a 500 or 503 error message. The assessment requires that a number of changes are made to be messages before they are sent to a Mock CDR system (provided). Selected messages need to be saved to disk, which will be submitted along with your Rhapsody configuration for the assessment.

All the information required to complete the configuration for the final assessment can be found in the lessons and exercises you have already completed as part of the Rhapsody Professional course.

Tasks

The tasks to be performed to complete the Rhapsody Professional final assessment are summarized in the diagram below.



Final assessment tasks

Resources

The following resources are included to help you complete the assessment. The lookup table and Mock CDR are provided in a zip file that accompanies these instructions.

SQLite Database

The SQLite database required for the assessment is *Providers.db*. You may also require the SQLite JDBC database driver. This is included with the exercises in the "Working with Databases" module.

Lookup Table

The CSV file required to perform the diagnostic service section lookup is provided in *LabTestType.csv*.

Mock CDR System

The RLC containing the mock CDR system is included in *Mock CDR System.rlc*. By default, the system listens on port 12107. You may edit the RP_MockCDR_TCPPort Rhapsody Variable to change the port number the configuration listens on if required. Do not attempt to edit any other parts of the configuration, as this will likely result in a solution that does not work correctly when tested.

Do not include the Mock CDR in the configuration you submit for the final assessment.

Message Validator

You can test the messages generated by your configuration for correctness using the message validator.

We recommend that you test a large selection of messages generated by your configuration. This will help identify any errors in your configuration before submitting it.

Exercise Tasks

To complete the final assessment, the following set of tasks must be completed using Rhapsody.

It must be possible for your solution to run for an extended period of time without sending any messages to the error queue. The solution should include configuration to check for errors that might occur as part of normal operations. Make sure the configuration is adequately documented to describe the "what" and "why".

Receive Messages

A request must be made to the following web service to retrieve new laboratory results once every 30 seconds.

<http://resources.edu.rhapsody.global/professional/assessment/getReports?user=<email>>

Where <email> is the email address associated with your Academy account.

Responses from the service must be processed based on the HTTP response code as follows:

Response Code	Action
200	Process according to all remaining assignment instructions and send to the Mock CDR system.
<i>All response codes between 500 and 599</i>	Send the XML message to a directory for review. Files must use the /response/reference field from the XML as the filename and have a .error filename suffix.
<i>All other response codes</i>	Delete.

Extract Reports From Message

Each XML message will contain one or more reports. Each report in the received message that contains an order must be processed separately though the remaining steps in the assessment.

Map Reports to HL7

All reports must be mapped to an HL7 version 2.4 ORU^R01 message. The table below details the mapping rules that should be used when constructing the HL7 message.

Unless otherwise stated, all repeats of an input field need to be mapped to their output equivalent.

Input XML XPath	Output HL7 Field	Notes
	MSH.SendingApplication.Nam espaceID	Use contents of X- Powered-By HTTP response header.

/response/report/site/location	MSH.SendingFacility.NamespaceID	
	MSH.DateTimeOfMessage	Use the current date time with a second precision.
	MSH.MessageControlID	Use JavaScript Math.random()(opens in a new tab) function.
	MSH.ProcessingID.ProcessingID	Hard code with a value of "T".
/response/report/patient/identifier	PID.PatientIdentifierList.IDNumber	Map when the @type attribute has a value of "MR".
/response/report/patient/identifier/@type	PID.PatientIdentifierList.IdentifierTypeCode	Map when the @type attribute has a value of "MR".
/response/report/patient/name/familyName	PID.PatientName.FamilyName.Surname	
/response/report/patient/name/givenName[1]	PID.PatientName.GivenName	
/response/report/patient/name/givenName[2]	PID.PatientName.MiddleInitialOrName	
/response/report/patient/dateOfBirth	PID.DateTimeOfBirth.Time	Map date to HL7 date format.
/response/report/patient/sex	PID.Sex	First character of field in XML matches equivalent HL7 code.
/response/report/order/orderNumber [@system = 'lab']	OBR.PlacerOrderNumber.EntityIdentifier	

/response/report/order/orderNumber [@system = 'pas']	OBR.FillerOrderNumber.EntityIdentifier	
/response/report/order/service/identifier	OBR.UniversalServiceID.Identifier	
/response/report/order/service/description	OBR.UniversalServiceID.Text	
/response/report/order/observationTime	OBR.ObservationDateTime.Time	Map date to HL7 date format.
/response/report/order/reportTime	OBR.ResultsRptStatusChngDateTime.Time	Map date to HL7 date format.
/response/report/order/result/observation/@format	OBX.ValueType	Map according to following rules: • Number > NM • <i>All other values</i> > ST
/response/report/order/result/test/identifier	OBX.ObservationIdentifier.Identifier	
/response/report/order/result/test/description	OBX.ObservationIdentifier.Text	
/response/report/order/result/observation	OBX.ObservationValue	
/response/report/order/result/units	OBX.Units.Identifier	
/response/report/order/result/referenceRange	OBX.ReferencesRange	
/response/report/order/result/abnormality	OBX.AbnormalFlags	First character of field in XML matches equivalent HL7 code.

/response/report/order/result/status	OBX.ObservationResultStatus	First character of field in XML matches equivalent HL7 code.
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Any fields in the XML message not mentioned in the mapping table above should be left unmapped. All dates in the incoming XML message are [ISO-8601\(opens in a new tab\)](#) formatted.

Database Lookup

The provider details must be updated in the message from the the provided SQLite database. The database contains a Practitioner table that contains provider details. The Identifier column in the table is used to match the contents of the /response/report/order/provider field in the incoming message.

The following fields in the message should be updated with the contents of any record returned from the database:

Database Field	Output HL7 Field	Notes
familyName	OBR.OrderingProvider.FamilyName	
givenName	OBR.OrderingProvider.GivenName	
middleName	OBR.OrderingProvider.MiddleInitialOrName	
prefix	OBR.OrderingProvider.Prefix	

All messages must continue processing after the database lookup has completed.

Terminology Web Service

You need to create a SOAP web service that maps a universal service identifier to its related diagnostic service section identifier. The web service will accept a request containing a single parameter of universal service id. The response message will contain a single parameter containing the associated diagnostic service section id. The CSV file provided as part of the assessment resources must be used to perform the mapping between the universal service id and the diagnostic server section id.

Terminology Lookup

A call must be made to the terminology web service created above to set the OBR.24 (diagnostic service section id) field in the generated HL7 message.

The /message/report/order/service/identifier field from the incoming messages is used to perform the query. If the web service is unable to find a mapping for the incoming universal service id, the diagnostic service id should be left blank in the message. All messages must continue processing after the terminology lookup has completed.

Send to Mock CDR System

Included in the resources for the final assessment is a Rhapsody RLC that mimics the behavior of a CDR system. All received XML messages containing a report must be converted to HL7 and updated and then sent to this system. A Message Tracking scheme must be used to track each message sent to the CDR.

Message Tracking Scheme

All messages sent to the mock CDR system must be monitored by a message tracking scheme with the following rules:

- Resend messages which either generate an AE acknowledgement or do not return an acknowledgement message.
- Save messages which generate an AR acknowledgement to a directory. Files must include the placer order number from the original message as part of the filename and have an .ar filename suffix.

In the event of a tracking error, subsequent messages must continue to be processed.

Report on Abnormal Results

All ORU^R01 messages that have been successfully acknowledged by the Mock CDR system with a code of AA must be reported on by being saved to a directory if at least one OBX segment contains an abnormality.

Messages are considered abnormal when the OBX.8 field:

- Has a value of other than N, and
- Is not an empty field.

Files must include a patient identifier and the placer order number from the message as part of the filename, and have a .report filename suffix.

Messages that do not contain abnormal results do not require any further processing.

Reviewing Your Solution

Configuration Checklist

The following checklist can be used to confirm that your submission meets all assessment requirements.

Checklist	Description of the Requirements	If not met
Items included in .rlc	All configuration needed to run solution is included. • Solution (routes, communication points) • Message definitions - explicitly named (.s3d, .xsd) • Mapper definition - explicitly named (.mdf) • Message tracking scheme • SOAP web service • Lookup table • Rhapsody variables	Automatic resubmission
Routes	• Well-designed layout • Functional separation of routes • Routes well grouped / well named folders • Naming convention (components / conditional connectors) • Detailed route documentation (standard template used – author, date, purpose, changelog)	Potential resubmission
Database lookup	Correct data returned and correct provider fields updated.	Automatic resubmission
SQL	SQL code well-structured: • Newlines and indentation used to improve readability.	Lower grade
Map functionality	• Submaps structured to follow the output message definition. • Comments in mapping code (code comments, author, date, purpose, changelog).	Potential resubmission
Lookup table	• Table well named and has description. • Table correctly defined and loaded with supplied data from CSV file.	Potential resubmission

Web service hosting	• Sensible web service Name, Namespace, and Description. • Sensible operation names used. • Web service accepts single universal service id parameter and responds with single diagnostic service sector id parameter. • Web service hosting lookup (in a separate route).	Potential resubmission
Web service client	• Web service correctly called. • Web service response always correct and matches with correct message. • OBR.24 field is correctly updated on successful lookup and not updated on lookup failure.	Potential resubmission
JavaScript	• JavaScript code well-structured. • Liberal comments to describe purpose of JavaScript code.	Potential resubmission
All Messages sent to mock CDR	• Messages sent to mock CDR monitored by explicit message tracking scheme. • Response handling: ○ Messages that generate an AE acknowledgement must be resent. ○ Messages that generate an AR acknowledgement must be saved to a directory.	Automatic resubmission
Check for abnormal messages	Only abnormal messages correctly filtered.	Automatic resubmission
Message Validator	All message types (including: .error, .ar, & .report files) generated by the solution can be processed by the Message Validator (opens in a new tab) without error.	Automatic resubmission

Error handling	- No messages are inadvertently sent to the error queue during testing. - Appropriate error checking added to configuration to detect and process errors that might occur during processing.	Potential resubmission
Use of Rhapsody variables	All parameters that could be expected to change between two environments where the configuration could be deployed should be a variable.	Lower grade

Rhapsody Best Practice

It is expected that your solution will demonstrate an understanding of the best practices discussed in the Rhapsody Professional course. During assessment of your submission, a grade based on conformance to best practices will be assigned. The grade assessment includes consideration of SQL, JavaScript, and Mapper formatting along with route layout, component naming, error handling, and documentation (both notes and support notes).

We suggest that you carefully test your solution prior to submission to ensure that all messages are processed correctly. This should include Filter Testing individual components in your solution, end-to-end testing of the complete solution, and use of the message validator service.

ll final assessment submissions are independently graded. The grade awarded to an assessment is based on the following matrix:

Grade	Overview	Detailed Description
A	Exemplary solution	An excellent solution that meets all assessment requirements and fully demonstrates the use of Rhapsody's Best Practice principles.

Grade	Overview	Detailed Description
B	Solid solution, appropriate to implement on site	A working solution that processes messages in the required way. The assessment comments are likely to include one or more recommendations for improvements to its configuration.
C	Solution works, but is difficult to maintain and update	A working solution, but having significant issues requiring attention before implementation is possible in a Production environment. Resubmission request likely.
Not Achieved	Messages are not being processed in accordance with the assessment instructions	One or more of the assessment requirements have not been met and there are many deviations from Rhapsody's Best Practice principles. Resubmission requested.

Resubmitting your Solution

When a resubmission is requested, the assessor will provide a detailed list of the changes required to bring your solution up to standard. Please ensure these suggestions are implemented in the new files you submit. You have three attempts to pass.

If you have any queries regarding your resubmission, please reply directly to your assessor. He or she will be able to answer questions about the grading of your solution.

The following files must be submitted at the end of this assignment for assessment:

- The PDF documentation of your Rhapsody configuration.
- The Rhapsody RLC containing your final assessment configuration.

- Several files containing error messages from the laboratory service.
- Several files (preferably more than 20) containing messages that generated an AR acknowledgement.
- Several files (preferably more than 20) containing abnormal messages.

Save all files into a **Zip** Archive and submit this to the **Final Assessment File Upload**.

① We use the submitted sample files to check that your solution is working correctly. We may ask you to send us additional output files to help us validate your solution.

Uploading Your Solution

Your Zip file is uploaded to the **Academy** using the **Final Assessment File Upload** link on the first page of this assessment module. This link will become available once you have finished reading all the final assessment instructions.

Your assessment solution will be graded once it has been uploaded to the Academy and the submission submitted.